

SAT 8

Math Module 3

Question #	ID		Answer
SAT8 M 3.1	e7b6f0d1	$4x + 6 = 18$ Which equation has the same solution as the given equation? A. $4x = 108$ B. $4x = 24$ C. $4x = 12$ D. $4x = 3$	C
SAT8 M 3.2	88e13c8c	The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is \$400? A. \$13,400 B. \$13,000 C. \$15,400 D. \$37,400	C
SAT8 M 3.3	a08ma303	If the graph of $27x + 33y = 297$ is shifted down 5 units in the xy-plane, what is the y-intercept of the resulting graph? (A) (0, 4) (B) (0, 6) (C) (0, 14) (D) (0, 28)	A
SAT8 M 3.4	aad7e1b9	The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane? A. $(-2, 0)$ B. $(0, -2)$ C. $(0, \frac{1}{10})$ D. $(\frac{1}{10}, 0)$	B

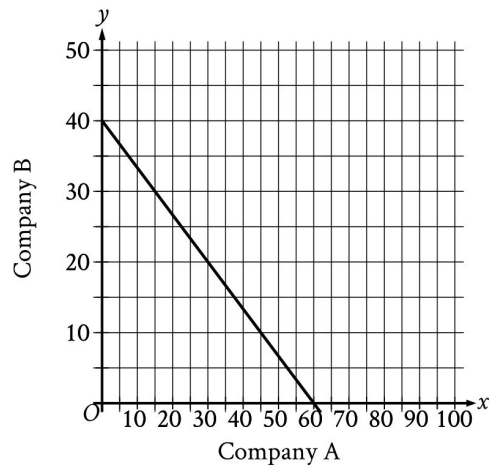
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Question # ID
SAT8 M 3.5 2e0290c3

Answer

B



The graph shows the relationship between the number of shares of stock from Company A, x , and the number of shares of stock from Company B, y , that Simone can purchase. Which equation could represent this relationship?

- A. $y = 8x + 12$
- B. $8x + 12y = 480$
- C. $y = 12x + 8$
- D. $12x + 8y = 480$

SAT8 M 3.6 a08ma306

$$\begin{aligned}\frac{7}{2}x + 6y &= 25 \\ \frac{5}{2}x + 6y &= 23\end{aligned}$$

D

The solution to the given system of equations is (x, y) . What is the value of $\frac{17}{2}x + 18y$?

(A) 2

(B) 3

(C) 48

(D) 71

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Question # ID
SAT8 M 3.7 6ecdbcb4

Answer

B

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

SAT8 M 3.8 1853bb35

C

For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

A. $q(x) = 0.55(14)^x$

B. $q(x) = 1.45(14)^x$

C. $q(x) = 14(0.55)^x$

D. $q(x) = 14(1.45)^x$

SAT8 M 3.9 a08ma309

35

A submersible device is used for ocean research. The function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

Question # ID
SAT8 M 3.10 9966235e

A

A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

A. 149,796

B. 164,500

C. 190,955

D. 310,800

SAT8 M 3.11 158591f0

29/3

$$x(x + 1) - 56 = 4x(x - 7)$$

What is the sum of the solutions to the given equation?

SAT8 M 3.12 d8539e09

C

$$y < 6x + 2$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

- A.
- | | |
|-----|-----|
| x | y |
| 3 | 20 |
| 5 | 32 |
| 7 | 44 |
- B.
- | | |
|-----|-----|
| x | y |
| 3 | 16 |
| 5 | 36 |
| 7 | 40 |
- C.
- | | |
|-----|-----|
| x | y |
| 3 | 16 |
| 5 | 28 |
| 7 | 40 |
- D.
- | | |
|-----|-----|
| x | y |
| 3 | 24 |
| 5 | 36 |
| 7 | 48 |

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Question # ID
SAT8 M 3.13 b0a72bdc

Answer

B

What is the diameter of the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

- A. 4
- B. 8
- C. 16
- D. 32

SAT8 M 3.14 1ce9ffcd

C

$$-9x^2 + 30x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. 3
- B. 0
- C. -25
- D. -53

SAT8 M 3.15 a08ma315

338

In triangle XYZ , angle Y is a right angle, the measure of angle Z is 33° , and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant, what is the value of k ?

SAT8 M 3.16 040f2a84

1.8

The regular price of a shirt at a store is \$11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the \$ sign when entering your answer. For example, if your answer is \$4.97, enter 4.97)

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Question # ID
SAT8 M 3.17 901c3215

Answer

C

In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

- A. The measure of angle A
- B. The length of side AB
- C. The lengths of sides BC and EF
- D. No additional information is necessary.

SAT8 M 3.18 45a534d0

-34

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

SAT8 M 3.19 a08ma319

D

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

(A) $3x^2 + 7x + 14b$

(B) $3x^2 + 28x + 14b$

(C) $3x^2 + 42x + 14b$

(D) $3x^2 + 49x + 14b$

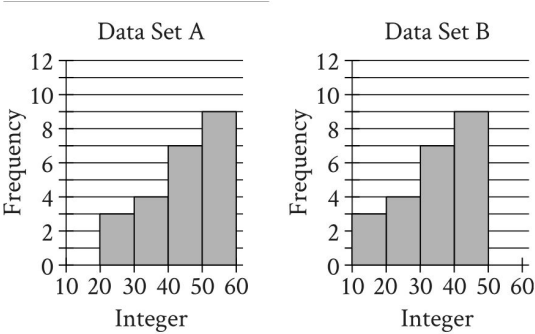
SAT8 M 3.20 a4bd60a3

104

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

Question # ID
SAT8 M 3.21 f8a322d9

Answer
B



Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 23

SAT8 M 3.22 916ffe9b

B

Poll Results	
Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

- A. 163
- B. 1,304
- C. 3,864
- D. 5,621